STAT 4620 Final Project Report

Daphne Kaur, Arjun Narang, Charlene Li, and Samhit Kasichainula

1. Introduction

Our goal is to model the relationship between workers' wages and a set of demographic and labor-market characteristics using the Wage dataset from the ISLR package. Understanding how factors such as age, education, job classification, year, and marital status contribute to wage outcomes is important for both labor economics research and practical workforce planning. The objective is to identify a model that is both interpretable and well-supported by the data, and to summarize insights about which variables most strongly influence wages and how these effects behave across demographic groups.

2. Data Description and Cleaning

Dataset Structure

The Wage dataset includes 3000 observations with 13 variables. Table 1 shows the description and measurement type for each variable.

Table 1: Variable Descriptions for Wage Dataset

Variable	Description	Type
X	Respondent ID (unique identifier)	Numerical: Discrete
year	Year that wage information was recorded	Numerical: Discrete
age	Age of worker	Numerical: Discrete
maritl	Marital status: 1. Never Married, 2. Married, 3. Widowed, 4. Divorced, 5. Separated	Categorical: Nominal
race	Race of worker: 1. White, 2. Black, 3. Asian, 4. Other	Categorical: Nominal
education	Highest education level: 1. < HS Grad, 2. HS Grad, 3. Some College, 4. College Grad, 5. Advanced Degree	Categorical: Ordinal
region	Region of country (mid-atlantic only)	Categorical: Nominal
jobclass	Job class: 1. Industrial, 2. Information	Categorical: Nominal
health	Self-reported health status: 1. <=Good, 2. >=Very Good	Categorical: Ordinal
health_ins	Health insurance status: 1. Yes, 2. No	Categorical: Binary
logwage	Log-transformed workers wage	Numerical: Continuous
wage	Workers raw wage	Numerical: Continuous
Resp	Mystery response variable	Numerical: Continuous

Data Cleaning & Preprocessing

All original variables are complete. The appended variable Resp contains missing values. To examine potential bias, we created an indicator variable marking whether Resp was missing. Comparing summary statistics of numerical predictors between missing and non-missing Resp cases showed similar mean values, suggesting that the missingness is likely missing at random (Table 2). Therefore, removing rows with missing Resp will not bias the analysis. We also checked for duplicate rows and found none. Thus, our final cleaned dataset excludes the identifier variable X and all rows with missing Resp.

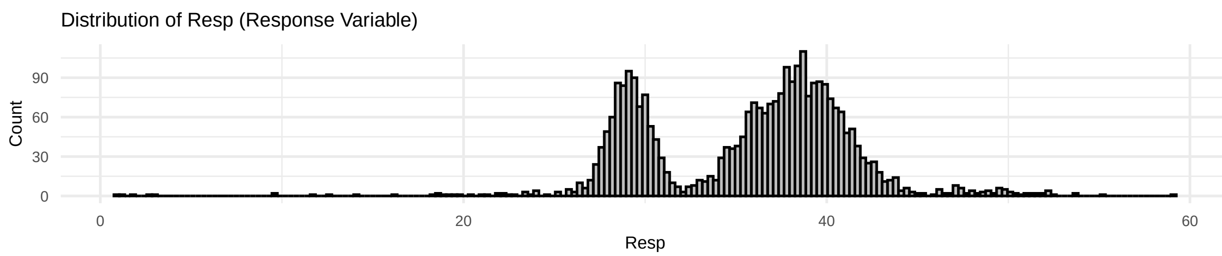
Table 2: Mean of Numeric Variables by Missingness of Resp

missing	X	year	age	logwage	wage	Resp
FALSE	219006.4	2005.79	42.42	4.65	111.73	35.66
TRUE	212854.1	2005.62	42.15	4.67	110.65	NaN

3. Exploratory Data Analysis

3.1 Distribution of the Response Variable (Resp)

Before examining predictors, we examine the shape of the response variable. The histogram below shows that **Resp is clearly bimodal**, with two distinct peaks. The strong **bimodal distribution** suggests that a linear mean structure may not fully capture the data, and some predictors may shift individuals into one mode or the other. This motivates the need for polynomial and semi-parametric models.



3.2 Numeric Variables & Categorical Variables

The variable wage shows a wide range, with a median that is lower than the mean, indicating the presence of some high earners. The variable logwage is the logarithmic transformation of wage, which reduces skewness and makes the distribution more symmetric (Table 3). The categorical variables were summarized by their most frequent categories (Table 4).

Table 3: Summary Statistics – Numeric Variables

Variable	Min	Median	Mean	Max
year	2003.00	2006.00	2005.79	2009.00
age	18.00	42.00	42.42	80.00
logwage	3.00	4.65	4.65	5.76
wage	20.09	104.92	111.73	318.34
Resp	1.00	36.95	35.66	59.11

Table 4: Most Frequent Category – Categorical Variables

Variable	Most Frequent Category
maritl	2. Married
race	1. White
education	2. HS Grad
region	2. Middle Atlantic
jobclass	1. Industrial
health	2. >=Very Good
health_ins	1. Yes

Correlation Analysis (Appendix 1)

The variables wage and logwage are highly correlated, as expected due to the log transformation. The response variable Resp shows moderate positive correlations with wage, logwage, and age, suggesting that these variables may be influential in predicting Resp. The year variable shows a weak correlation with Resp, indicating limited time-based effects.

Numeric Variables vs Response (Appendix 2)

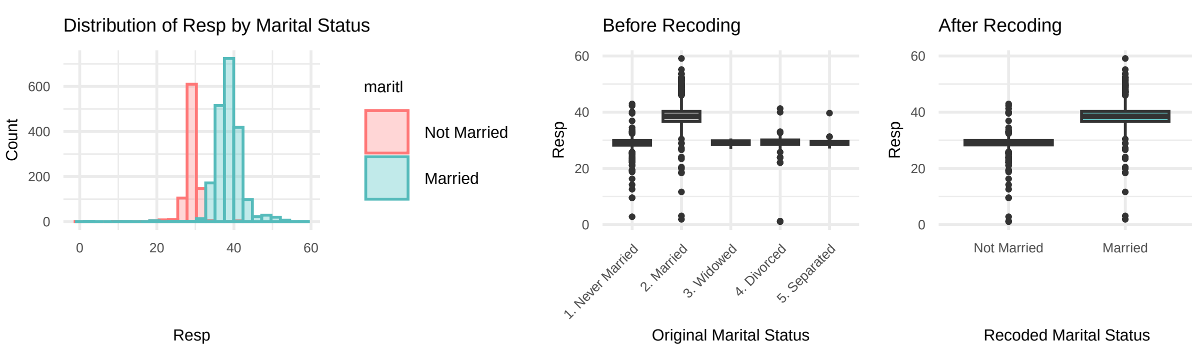
The scatterplot matrix shows a positive, non-linear relationship: wages increase with age up to around 60, then level off or slightly decrease, with variance also growing with age. **Resp is highly correlated with wage and logwage and positively associated with age**. The clusters in the age versus Resp plot likely result from a strong categorical predictor, which can create distinct bands when plotted against a continuous variable. Similarly, the wage and logwage versus Resp plots show clusters influenced by the same categorical predictor.

Categorical Variables vs Response (Appendix 3)

- education: Clear, strong separation between levels. Higher education corresponds to higher Resp.
- maritl (Married vs Unmarried): Marital status shows one of the strongest relationships in the data, with married individuals disproportionately appearing in the upper mode of the Resp distribution. This pattern likely contributes to the bimodal distribution shape.
- race, health, health_ins: These show weaker group separation. Their effects appear smaller and do not meaningfully explain the bimodality of Resp.
- jobclass: Practically no separation in Resp; likely not an important predictor.
- Outlier Identification: The outliers show data points with unusually low or high wages for their respective categories, which may warrant further investigation for potential influential observations.

3.3 Recoding Marital Status (Married vs. Unmarried)

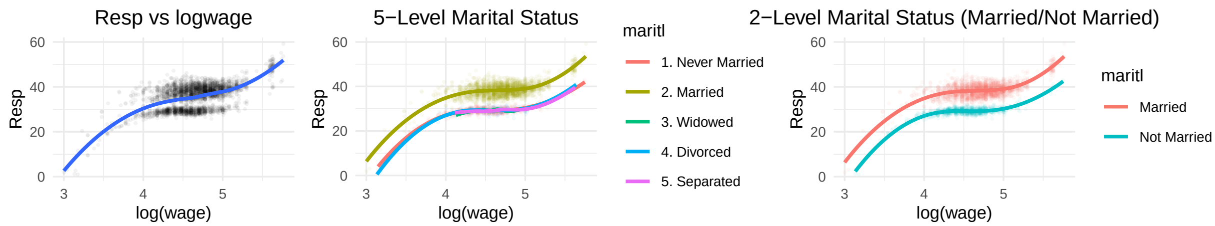
The boxplots below show a clear distinction between the married and unmarried classes of the maritl variable in relationship to the response variable. After coding the maritl variable as binary (1 = Married, 0 = Unmarried), plotting the response and coding it according to the new binary variable, we see that the the two distinct modes in the bimodal response appear to be explained by marital status. Therefore, for the remainder of this report, maritl will be modeled as a binary variable.



3.4 Understanding Interactions

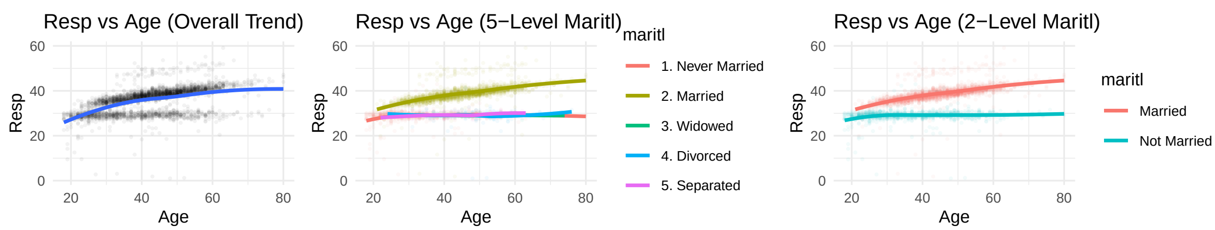
Resp vs log(Wage)

While the model without marital status shows one smooth upward trend, the inclusion of marital status produces two distinct trajectories, especially at higher wage levels, illustrating a meaningful interaction effect.



Resp vs Age

Including marital status reveals an interaction effect, where married individuals display consistently higher and more rapidly increasing response values across age.



4. Modeling Strategy and Results

4.1 Data Partitioning and Modeling Approach

The cleaned data ($n = 2,940$) was partitioned into an **80% training set** (Wage.train) for model fitting and hyperparameter tuning, and a **20% test set** (Wage.test) for unbiased performance evaluation.Our modeling strategy moved from simple parametric models to complex non-parametric methods to find the optimal balance between predictive accuracy (low Test MSE) and statistical inference (interpretability):

- **Interpretable Non-Linear Models:** (Polynomial Regression, GAMs) capture non-linear trends and key interactions.
- **Non-Parametric Ensemble Models:** (Random Forest, Boosting) offer maximum predictive power by capturing arbitrary interactions.

4.2 Interpretable Non-Linear Models

4.2.1 Polynomial Regression:

Polynomial regression was first used as a simple and interpretable way to introduce curvature into the model, capturing **nonlinear patterns** while **preserving a largely linear structure** and serving as a **baseline for assessing nonlinearity**. Models were fit both with and without interaction terms for marital status to evaluate whether these interactions improved accuracy. The best-performing polynomial model used a third-degree polynomial for log(wage) and a second-degree polynomial for age, with both terms fully interacting with marital status (recoded as Married/Not Married), along with all other categorical predictors.

$$\text{Resp} \sim \text{poly}(\text{logwage}, 3) \times \text{maritl} + \text{poly}(\text{age}, 2) \times \text{maritl} + \text{All Categoricals}$$

Table 5: log(Wage) and Age Polynomial Test MSE Results

Model	Test_MSE
Age Polynomial (Degree 2)	23.124524
log(Wage) Polynomial (Degree 3)	19.075852
Age Polynomial (Degree 2) with Marriage Interaction	8.068780
log(Wage) Polynomial (Degree 3) with Marriage Interaction	4.043070
Age and log(wage) Polynomial (no interaction)	14.669246
Age and log(wage) with marital status interaction	1.035746

The results show that adding nonlinear terms for both predictors does not improve performance by much unless interaction effects are included, indicating that subgroup differences by marital status drive much of the predictive gain.

Introducing Categorical Predictors

In an effort to improve the model's performance, the age and log(Wage) model with interaction effects is set as the baseline. From there, various categorical values were tested to gauge an improvement in Test MSE. The top-performing polynomial models all include the full or near-full set of categorical predictors, suggesting that demographic and job-related characteristics contribute additional explanatory value beyond nonlinear age and wage effects.

Table 6: Top 5 Polynomial Models by Test MSE

Model	Test_MSE
Base + education + race + jobclass + health + health_ins	0.976
Base + education + jobclass + health + health_ins	0.977
Base + education + race + jobclass + health_ins	0.977
Base + education + jobclass + health_ins	0.978
Base + education + race + jobclass + health	0.979

4.2.2 Generalized Additive Models (GAM):

GAMs were introduced to address the limitations of polynomial models by learning smooth, data-driven functions rather than relying on fixed polynomial shapes, allowing flexible nonlinear modeling while maintaining interpretability. They also permit different smooths by marital status, aligning with previously observed interaction effects, and the model was later expanded to include additional categorical predictors such as education, race, and job class. A GAM was then fitted to replicate the baseline structure of the polynomial models using log(wage) and Age with their marital-status interactions. Unlike polynomial regression, **GAM uses smoothing splines to provide a flexible, non-parametric fit that better captures nonlinear patterns without imposing a global functional form**. The **best model** employed smoothing splines for log(wage) and age, with smooth functions varying by marital status (referred to as gam12 in the internal analysis):

$$\text{Resp} \sim s(\text{logwage}, \text{by} = \text{maritl}) + s(\text{age}, \text{by} = \text{maritl}) + \text{All Categoricals}$$

Table 7: Top 5 GAM Models by Test MSE

Model	Test_MSE
Base + education + race + jobclass + health + health_ins	0.986
Base + education + jobclass + health + health_ins	0.987
Base + education + race + jobclass + health_ins	0.987
Base + education + jobclass + health_ins	0.988
Base + education + race + jobclass + health	0.989

Similar to polynomial regression, these results suggest that the GAM framework benefits from a full or nearly full list of additional categorical predictors, which enhance predictive accuracy when paired with spline-based nonlinear effects.

Comparison to Polynomial Regression

Compared to the polynomial regression models, the GAM produced slightly higher Test MSE, likely because its smoother spline-based functions allows for flexibility by attempting to capture curvature in the data. This slight decrease in bias does result in a slightly lower residual variance, as observed in Table 8. The best polynomial model achieves slightly lower test MSE (0.976 vs 0.986) and lower 10-fold CV MSE (1.031 vs 1.056) than the GAM, indicating marginally better out-of-sample performance and stability with fewer effective degrees of freedom (22 vs 31.11). Although the GAM attains a very slightly lower residual variance, this comes at the cost of greater complexity, so the polynomial model offers a better balance of accuracy, parsimony, and interpretability.

Table 8: Comparison of Best Polynomial and GAM Models

Metric	Best.Polynomial	Best.GAM
Test MSE	0.976	0.986
10-fold CV MSE	1.031	1.056
Residual Variance	1.013	1.010
Total Effective DF	22.000	31.110

4.3 Tree Based Method

Tree-based approaches were implemented to compare the performance of parametric models against non-parametric methods. CART, bagging, random forests, and boosting were fitted to the data, allowing for flexible modeling of the response variable.

Table 9: Comparison of Tree Based Methods

Model	Test MSE	CV MSE	Residual Variance	Effective DF
CART (Pruned)	3.528	4.214	3.428	8.000
Bagging	1.506	1.578	0.343	2234.716
Random Forest	1.676	1.583	0.476	2234.308
Boosting	1.390	1.565	1.212	9990.000

Looking at the table above, we can see that Boosting has the lowest Test MSE, but not the lowest CV MSE, residual variance, or effective degrees of freedom.

- **CV MSE: Boosting** has the **lowest (1.565)**, which is better than Bagging (1.578) and Random Forest (1.583).
- **Residual Variance: Bagging** and **Random Forest** have much lower values (0.343 and 0.476), indicating they fit the training data better than Boosting (1.212).
- **Effective Degrees of Freedom: CART (Pruned)** is by far the simplest, with only **8.000**, while **Boosting** is the most complex **(9990.000)**.

So, **Boosting** is best for predicting on new data (lowest Test MSE), but it is more complex and has higher training error than Bagging and Random Forest. The low residual variance and CV MSE for Bagging and Random Forest suggest they may be more stable and less prone to overfitting, but they don't generalize quite as well to the test set as Boosting in this case. CART, although by far the simplest, performed the worst among the four tree-based methods.

Pros and Cons of Tree Based methods:

Pros: Tree ensembles can automatically capture nonlinear and complex interactions without explicitly specifying polynomial terms or interactions, which is useful given the interplay between marital status, age, and log(wage) observed in the exploratory analysis. They handle mixed numeric and categorical predictors naturally and provide variable-importance summaries to identify key features driving predictions.

Cons: Despite this flexibility, tree-based models do not achieve the low Test MSE of the best polynomial model or the competitive performance of the GAM, offering no predictive advantage. Additionally, their effective degrees of freedom are vastly higher than those of the polynomial and GAM models, reflecting a far more complex fit without meaningful improvement in accuracy.

5. Conclusion and Best Model Selection

The final model selection is based on a rigorous comparison of the best-performing models across three families, evaluating both predictive accuracy (Test MSE/CV MSE) and model complexity/stability (Effective Degrees of Freedom and Residual Variance).

Table 10: Summary of Best Polynomial, GAM, and Tree-Based Models

Model	Test_MSE	10 fold CV MSE	Residual_Variance	Effective DF
Best Polynomial	0.976	1.031	1.013	22.00
Best GAM	0.986	1.056	1.010	31.11
Best Tree (Boosting)	1.390	1.565	1.212	9990.00

Analysis of Model Performance and Trade-offs:

1. Best Polynomial Model:

The best polynomial model shows the strongest predictive performance, with the **lowest Test MSE (0.976)** and **10-fold CV MSE (1.031)**, indicating accuracy and stability. It balances fit and complexity well, with low residual variance (1.013) and the **lowest Effective Degrees of Freedom (22.00)**, reflecting low bias and high parsimony. Its concise algebraic form also makes it the most interpretable, making it the preferred choice for statistical inference and clear communication.

2. Generalized Additive Model (GAM):

The GAM has slightly higher Test MSE (**0.986**) and CV MSE (**1.056**) than the polynomial model. Its smoothing functions offer more flexibility, giving a slightly lower residual variance (**1.010**) and better fit to the training data. However, this comes with greater complexity (Effective DF = **31.11**) without improving generalization, resulting in minor bias reduction but increased variance, making it a less optimal choice for prediction and practical use.

3. Best Tree-Based Model (Boosting):

The tree-based boosting model performs worse than the other approaches, with a Test MSE of **1.390**. It is highly complex (Effective DF = **9990.00**) and unstable, as shown by the large gap between its Test MSE and CV MSE (**1.565**), indicating severe overfitting. Its complexity and poor interpretability make it unsuitable for reliable prediction or meaningful statistical insights.

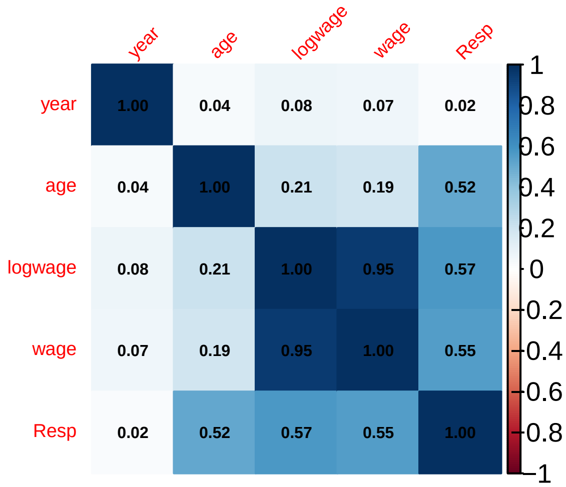
Final Conclusion:

The **Best Polynomial Model** remains the superior choice for this dataset.

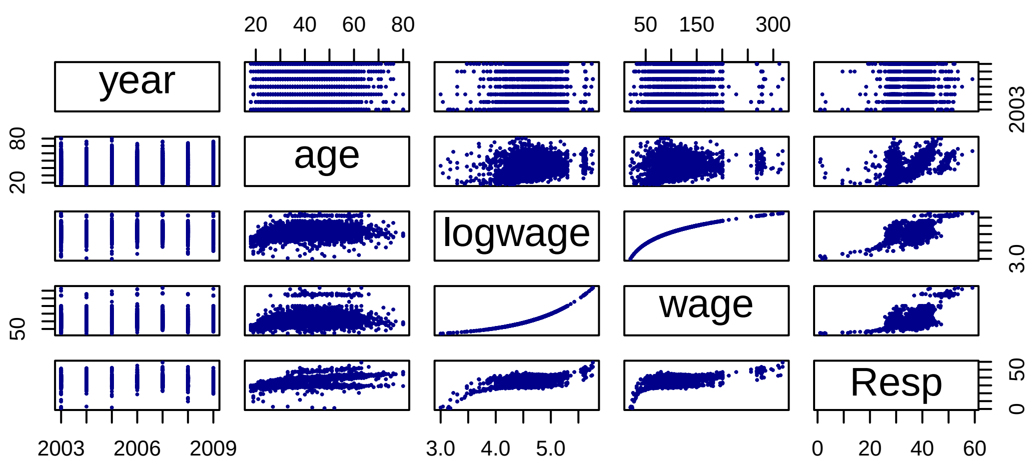
It clearly and demonstrably outperforms all other models in the primary metrics of interest (lowest Test MSE and lowest CV MSE) while maintaining the lowest Effective DF. The Polynomial Model's combination of high accuracy, stability, and interpretability offers the best overall solution, minimizing the risk of over-fitting and maximizing the potential for clear inference.

Appendix

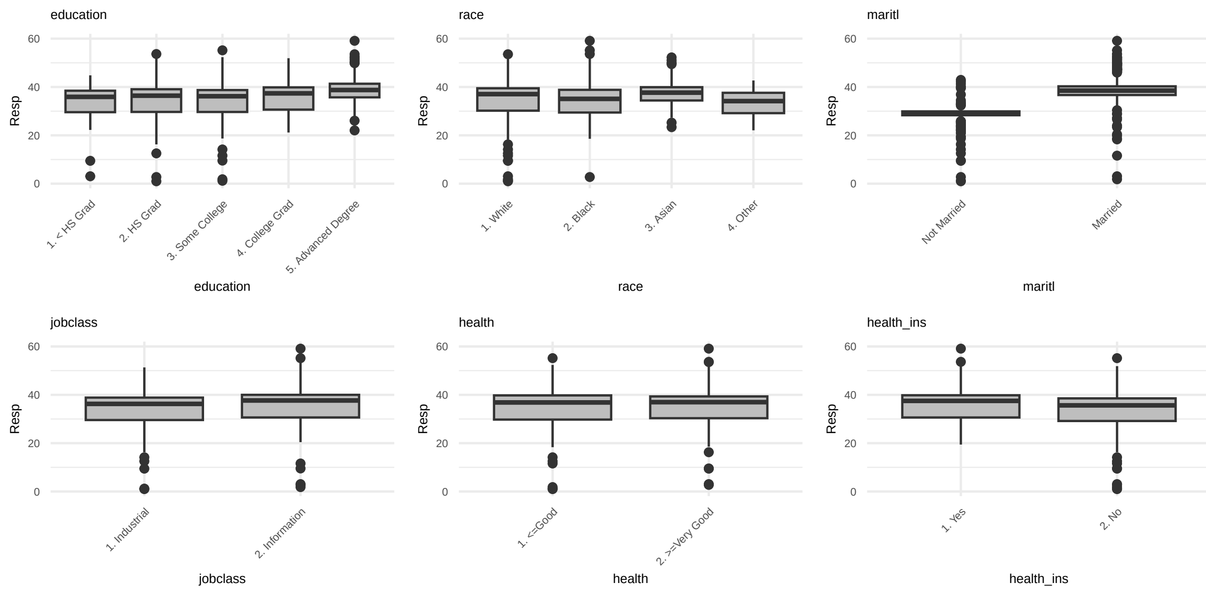
Appendix 1



Appendix 2



Appendix 3


education
Resp
60
40
20
0
1. < HS Grad
2. HS Grad
3. Some College
4. College Grad
5. Advanced Degree
education
race
1. White
2. Black
3. Asian
4. Other
race
maritl
Not Married
Married
maritl
jobclass
1. Industrial
2. Information
jobclass
health
1. <=Good
2. >=Very Good
health
health_ins
1. Yes
2. No
health_ins